Yi Ding

Ph.D. Student in Computer Science, Purdue University

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Education

Purdue University — IN, USA
Ph.D. in Computer Science — Aug. 2025 – Present
Advisor: Dr. Ruqi Zhang

Tianjin University — Tianjin, China
B.S. in Data Science and Big Data Technology — Aug. 2021 – Jun. 2025
School of Mathematics · Advisors: Dr. Bing Cao and Dr. Qinghua Hu

Research Interests

I study how large language and vision-language models can improve from limited, noisy, or imperfect supervision. My work develops reliable and efficient post-training algorithms for reasoning and safety, with a focus on reinforcement learning, on-policy distillation, and self-correction.

Selected Publications

* Equal contribution.

Reinforcement Learning & Self-Improvement

[S1] **Does On-Policy Distillation Really Distill? From Noisy Teacher to Self-Improvement**
Yi Ding, Ruqi Zhang
Preprint 2026 [PDF] [Project]
#1 on Hugging Face Daily Papers (Sep. 1, 2026)

[S2] **Learning Self-Correction in Vision–Language Models via Rollout Augmentation**
Yi Ding, Ziliang Qiu, Bolian Li, Ruqi Zhang
ICML 2026 [PDF] [Project]

[S3] **Sherlock: Self-Correcting Reasoning in Vision-Language Models**
Yi Ding, Ruqi Zhang
NeurIPS 2025 [PDF] [Project]

Multimodal Safety & Alignment

[S4] **Rethinking Bottlenecks in Safety Fine-Tuning of Vision Language Models**
Yi Ding*, Lijun Li*, Bing Cao, Jing Shao
ICLR 2026 [PDF] [Project]

[S5] **SafeWork-R1: Coevolving Safety and Intelligence under the AI-45° Law**
Shanghai AI Lab: Yicheng Bao, …, **Yi Ding**, … (100+ authors)
Technical Report 2025 [PDF]

[S6] **Visual Contextual Attack: Jailbreaking MLLMs with Image-Driven Context Injection**
Ziqi Miao*, **Yi Ding***, Lijun Li, Jing Shao
EMNLP 2025 (Main) [PDF]

[S7] **ETA: Evaluating Then Aligning Safety of Vision Language Models at Inference Time**
Yi Ding, Bolian Li, Ruqi Zhang
ICLR 2025 [PDF] [Project]

Additional Publications

* Equal contribution.

[A1] **Addressing Performance Saturation for LLM RL via Precise Entropy Curve Control**
Bolian Li, Yifan Wang, **Yi Ding**, Anamika Lochab, Ananth Grama, Ruqi Zhang
Preprint 2026 [PDF] [Project]

[A2] **Position: Modular Safety Guardrails Are Necessary for Foundation-Model-Enabled Robots in the Real World**
Joonkyung Kim, Wenxi Chen, Davood Soleymanzadeh, **Yi Ding**, Xiangbo Gao, Zhengzhong Tu, Ruqi Zhang, Fan Fei, Sushant Veer, Yiwei Lyu, Minghui Zheng, Yan Gu
ICML 2026 (Position Paper) [PDF]

[A3] **Test-Time Dynamic Image Fusion**
Bing Cao, Yinan Xia*, **Yi Ding***, Changqing Zhang, Qinghua Hu
NeurIPS 2024 [PDF]

[A4] **Predictive Dynamic Fusion**
Bing Cao, Yinan Xia*, **Yi Ding***, Changqing Zhang, Qinghua Hu
ICML 2024 [PDF]

Research Experience

RZ-Lab, Purdue University — IN, USA
Research Assistant — May 2024 – May 2025
Advisor: Dr. Ruqi Zhang

Open Trust Lab, Shanghai AI Laboratory — Beijing, China
Research Scientist — Dec. 2024 – Mar. 2025
Advisor: Dr. Lijun Li

MLDM Lab, Tianjin University — Tianjin, China
Research Assistant — Sep. 2023 – Dec. 2024
Advisors: Dr. Bing Cao and Dr. Qinghua Hu

Awards and Research Support

NSF ACCESS Discover Project Award — 2025–2026
Project: LLM Post-Training

Academic Service

Reviewer: ICLR 2025–2027; ICML 2026; NeurIPS 2025, 2026; CIKM 2026; ACL Rolling Review (May 2025).

Skills and Languages

Technical: Python, MATLAB, LaTeX; mathematical and statistical analysis.
Languages: Mandarin Chinese (native); English (TOEFL: 102).